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MSc Computer Science : Batch 2024-26, Semester I
DA104: Probability and Stochastic Process
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Answer the following questions

1. If for two events A and B, it so happens that $P(A) = \frac{3}{4}$ and $P(B) = \frac{3}{8}$, show that:
$$P(A \cup B) \geq \frac{3}{4} \text{ and } \frac{1}{8} \leq P(A \cap B) \leq \frac{3}{8}.$$
2. If the events A, B, and C are related as follows: $A \subset B \subset C$ and $P(A) = \frac{1}{4}$, $P(B) = \frac{5}{12}$, and $P(C) = \frac{7}{12}$, compute the probabilities of the following events:
$$A^c \cap B, A^c \cap C, A \cap B^c \cap C^c, A^c \cap B^c \cap C^c$$
3. A high school senior applies for admission to two colleges A and B, and suppose that: $P(\text{admitted at A}) = p_1$, $P(\text{rejected by B}) = p_2$, and $P(\text{rejected by at least one, A or B}) = p_3$. Calculate the probability that the student is admitted by at least one college; express it in terms of p_1, p_2, p_3.
4. Let E, F, and G be three events. Find expressions for the events so that, of E, F, and G,
   a) at least one of the events occurs;
   b) at least two of the events occur;
   c) all three events occur;
   d) exactly two of the events occur;
   e) at most two of the events occur;
   f) at most two of the events occur;
   g) at most three of the events occur.
5. If a die is rolled four times, what is the probability that six comes up at least once?
6. In a hand of bridge, find the probability that you have five spades and your partner has the remaining eight.
7. Suppose that the letters C, E, F, F, I, and O are written on six chips and placed into a box. Then the six chips are mixed and drawn one by one without replacement. What is the probability that the word "OFFICE" is formed?
8. A course in English composition is taken by 10 freshmen, 15 sophomores, 30 juniors, and 5 seniors. If 10 students are chosen at random, calculate the probability that this group will consist of 2 freshmen, 3 sophomores, 4 juniors, and 1 senior.
9. Two dice are thrown n times in succession. Compute the probability that double 6 appears at least once. How large need n be to make this probability at least $\frac{1}{2}$?
10. Find the probability that a number selected at random from the integers between 1 and $10,000$ inclusive is not divisible by $4, 6, 7, 10$.

11. Consider the following equation:

$$X_l + X_2 + X_3 + X_4 = 14$$

What is the probability that a random nonnegative integral solution will have the property that none of the x_i's exceed 8?

12. Compute the probability that a random permutation of $\{1, 2, ..., 8\}$ has no even integer in its natural position.

13. Compute the probability that a random permutation of $\{1, 2, ..., 8\}$ has exactly four integers in their natural positions.

14. At a party, seven gentlemen check their hats. If the hats are returned at random then compute the probability for the following events.

    a) None of the gentleman gets back their hat.

    b) At least one gentleman gets his hat.

    c) At least two of the gentlemen gets back his own hat.

15. Suppose in a box there are 3 chocobars, 4 apples and 2 bananas. You want to eat them by picking one item at random and consuming it. Compute the probability that you never ate all the items of same type consecutively.

16. Compute the probability that a random permutation $i_1, i_2, i_3, i_4, i_5, i_6$ of $\{1, 2, 3, 4, 5, 6\}$, satisfies the following constraint: $i_1 \neq 1, 5$; $i_3 \neq 2, 3, 5$; $i_4 \neq 4$; and $i_6 \neq 5, 6$.

17. A woman has n keys, of which one will open her door.

    a) If she tries the keys at random, discarding those that do not work, what is the probability that she will open the door on her k^{th} try?

    b) What if she does not discard previously tried keys?

18. Suppose that 5 of the numbers $1, 2, ..., 14$ are chosen. Find the probability that 9 is the third smallest value chosen.

19. If 8 people, consisting of 4 couples, are randomly arranged in a row, find the probability that no person is next to their partner.

20. Player A, B, C, D play bridge. Compute the probability that A's hand is void in at least one suit.